

Description

STM32MP157A/D

Table 1. STM32MP157A/D features and peripheral counts

Features		STM32MP157AADxx STM32MP157DADxx	STM32MP157AABxx STM32MP157DABxx	STM32MP157AACxx STM32MP157DACxx	STM32MP157AAAx STM32MP157DAAxx	Miscellaneous	
		TFBGA257	LFBGA354	TFBGA361	LFBGA448		
Package	Body size (mm)	10x10	16x16	12x12	18x18	-	
	Pitch (mm)	0.5 ⁽¹⁾	0.8	0.5 ⁽¹⁾	0.8		
	Ball size (mm)	0.30	0.40	0.30	0.40		
	Thickness (mm)	<1.2	<1.4	<1.2	<1.4		
	Ball count	257	354	361	448		
CPU		Dual-core Cortex-A7 FPU Neon TrustZone				-	
Caches size	2 × 32-Kbyte L1 data cache						
	2 × 32-Kbyte L1 instruction cache						
	256-Kbyte level 2 unified coherent cache						
Frequency		STM32MP157A: 2 × 650 MHz STM32MP157D: 2 × 800 MHz					
GPU		Vivante - Open GL ES 2.0				-	
for 3D graphics	Frequency	533 MHz					
MCU core		Cortex-M4 FPU				-	
	Frequency	209 MHz					
ROM		128 Kbytes (secure)				-	
Embedded SRAM		CPU system	256 Kbytes (securable)				708 Kbytes
		MCU subsystem	384 Kbytes				
		MCU retention	64 Kbytes				
		Backup	4 Kbytes (securable, tamper protected)				
SDRAM (securable)	LPDDR2/3	16-bit 533 MHz	Up to 1 Gbyte, single rank				-
		32-bit 533 MHz	-	-			
	DDR3/3L	16-bit 533 MHz	Up to 1 Gbyte, single rank				
		32-bit 533 MHz	-	-			
Backup registers		128 bytes (32x32-bit, securable, tamper protected)				-	

157A/D variant, diff mpu core speed

A : 650MHz

D : 800MHz

Table 1. STM32MP157A/D features and peripheral counts (continued)

Features			STM32MP157AADxx STM32MP157DADxx	STM32MP157AABxx STM32MP157DABxx	STM32MP157AACxx STM32MP157DACxx	STM32MP157AAAx STM32MP157DAAxx	Miscellaneous	
			TFBGA257	LFBGA354	TFBGA361	LFBGA448		
Timers	Advanced	16 bits	2				29 timers	
	General purpose	16 bits	8					
		32 bits	2					
	Basic	16 bits	2					
	Low power	16 bits	5					
	A7 timers	64 bits	2 × 4 (secure, non-secure, virtual, hypervisor)					
	M4 SysTick	24 bits	1					
	RTC/AWU		1 (securable)					
Watchdog			3 (independent, independent secure, window)				-	
Communication peripherals	SPI			6 (1 securable)			-	
		Having I2S		3				
	I2C (with SMB/PMB support)		6 (2 securable)				-	
	USART (smartcard, SPI, IrDA, LIN) + UART (IrDA, LIN)		4 + 4 (including 1 securable USART) some can be a boot source				Boot	
	SAI		4 (up to 8 audio channels), with I2S master/slave, PCM input, SPDIF-TX				-	
	USB	EHCI/OHCI Host		2 ports				-
				Embedded HS PHY with BCD				-
		OTG HS/FS (dual role port)		Yes, embedded FS or HS PHY with BCD, can be a boot source				Boot
				Embedded PHYs		3 (2 × high-speed + 1 × full-speed)		
	SPDIF-RX		4 inputs				-	
	FDCAN		2 (1 × TTCAN), clock calibration, 10 Kbyte shared buffer				-	
	HDMI-CEC		1				-	
	Including the following securable			1 × USART, 1 × SPI, 2 × I2C		1 × USART, 1 × SPI, 2 × I2C on securable GPIOs		-
SDMMC (SD, SDIO, eMMC)			3 (8 + 8 + 4 bits), eMMC or SD can be a boot source				Boot	
QuadSPI			Yes (dual-quad), can be a boot source				Boot	

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		TFBGA257	LFBGA354	TFBGA361	LFBGA448	
FMC	Parallel address/data 8/16-bit	-		4 × CS, up to 4 × 64 Mbyte		No boot
	Parallel AD-Mux 8/16-bit	4 × CS, up to 4 × 64 Mbytes				
	NAND 8/16-bit	Yes, 1 × CS, SLC, BCH4/8, can be a boot source				Boot
Gigabit Ethernet		-		MII, RMII, GMII, RGMII with PTP and EEE		-
10/100M Ethernet		MII, RMII with PTP and EEE				
LCD-TFT	Parallel interface	Up to 24-bit data, up to 90 MHz pixel clock (up to 1366 × 768 60 fps or up to 1920 × 1080 30 fps)				-
Display serial interface (DSI)		2 × data lanes 1 Gbps each (up to 1366 × 768 60 fps or up to 1920 × 1080 30 fps)				-
DMA		3 instances (1 securable), 48 physical channels in total				-
Hash		SHA-256, MD5, HMAC dual instances (secure and non-secure)				-
True random number generator		True-RNG, dual instances (secure and non-secure)				-
Fuses (one-time programmable)		3072 effective bits (secure, >1500 bits available for user)				-
Camera interface	Bus width	14-bit				-
GPIOs with interrupt (total count)		98		148	176	-
	Securable GPIOs	-		8		
	Wakeup pins	4		6		
	Tamper pins (active tamper)	2 (1)		3 (1)		
DFSDM		8 input channels with 6 filters				-
Up to 16-bit synchronized ADC		2 (up to 3.6/4.5/5/6 Msps on 16/14/12/10/8-bit each)				-
	Low noise 16 bit (differential)	-		2 (1)		
	16 bit (differential)	17 (7)		20 (9)		
	ADC channels in total ⁽²⁾	17		22		
12-bit DAC		2				-
Internal ADC/DAC VREF		1.5 V, 1.8 V, 2.048 V, 2.5 V or VREF+ input				-
	VREF+ input pin	Yes				

1. With inner matrix balls having 0.65 mm pitch to allow optimized PCB routing for supplies.

2. In addition, there is also 6 internal channels for temperature, internal voltage reference, V_{DDCORE} , $V_{BAT}/4$, DAC1 or DAC2 acquisitions.